



Air Condition Simulation Board

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.08-160815-01

1.0 Introduction

This simulation board was designed according to AC system of 140M Motor Grader. Training room capability is one of the very important keys to perform technical training in an efficient way. Typically, the Technician Career Development Program requires classroom and on-the-job training for each competency level. The classroom trainings give theoretical information for AC system of the machine and many trainees cannot imagine the functionality of the system that is written in the documentation.

Since the availability of a real air conditioning unit during classroom training is nearly impossible, the combination of both classroom and on-the-job training is very crucial.

The Simulation Board has been designed to find an optimum solution for cost savings and to combine classroom training with on-the-job training to deliver the efficient training sessions for trainees.

2.0 Best Practice Description

The air conditioning simulation board is designed to get similar electrical system and refrigerant circulation system of air conditioning system. The board consists of:

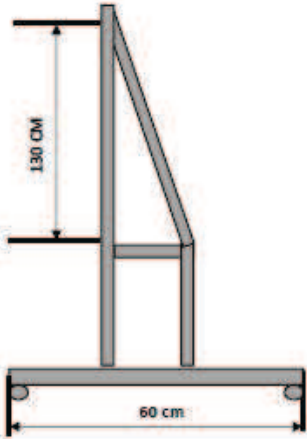
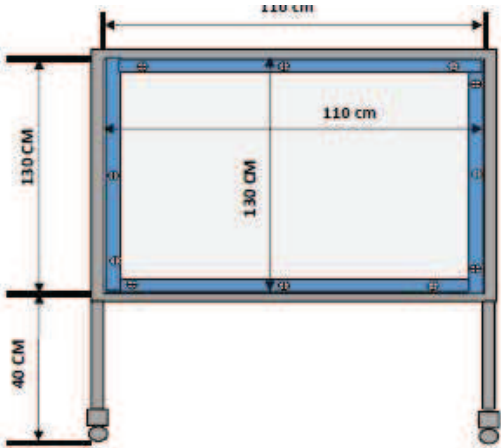
- ❖ Evaporator
- ❖ Condenser
- ❖ Orifice tube
- ❖ Inline dryer
- ❖ Accumulator
- ❖ Compressor
- ❖ Blower fan
- ❖ Water valve
- ❖ Air conditioner, heater control panel
- ❖ Blower motor resistor
- ❖ High and low pressure switches

Board function is like an air condition system. When the emergency switch is on. Air conditioner, heater control panel is energized and runs like real air conditioner electrical circuit. It allows trainees to check functionality, test the circuitry, measure voltage, current and resistance, visually understand the electrical flow priorities and detect the problem.

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3.0 Implementation Steps

Drawing and dimension of main body is following:



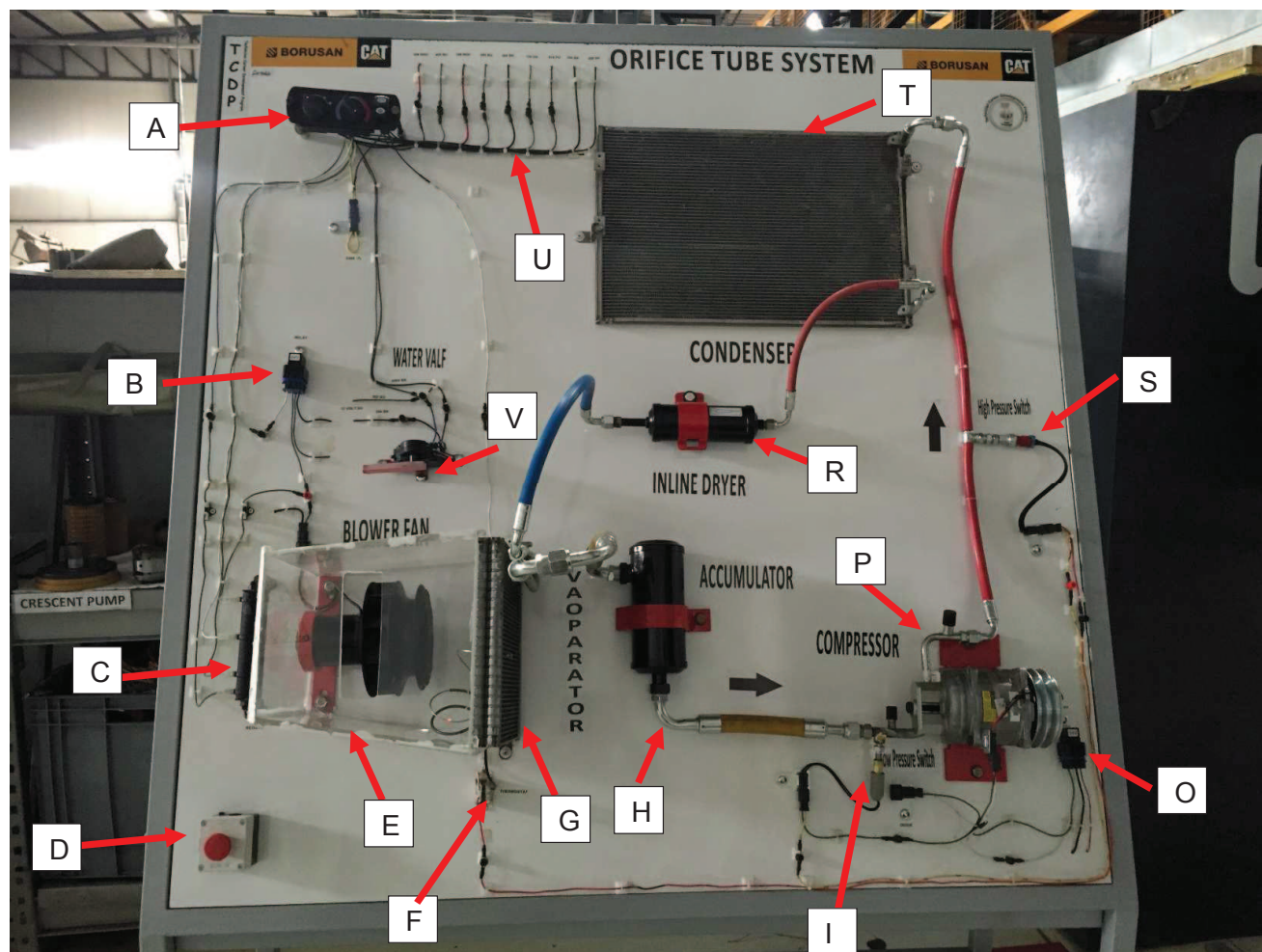
**AIR CONDITION
SIMULATION BOARD**

Drawing No	03	Draftsman	Salih Ertekin	 
Drawing name	Drawing of Board	Date	19.06.2016	

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The next step is to put the electrical ve AC component board and main body together.



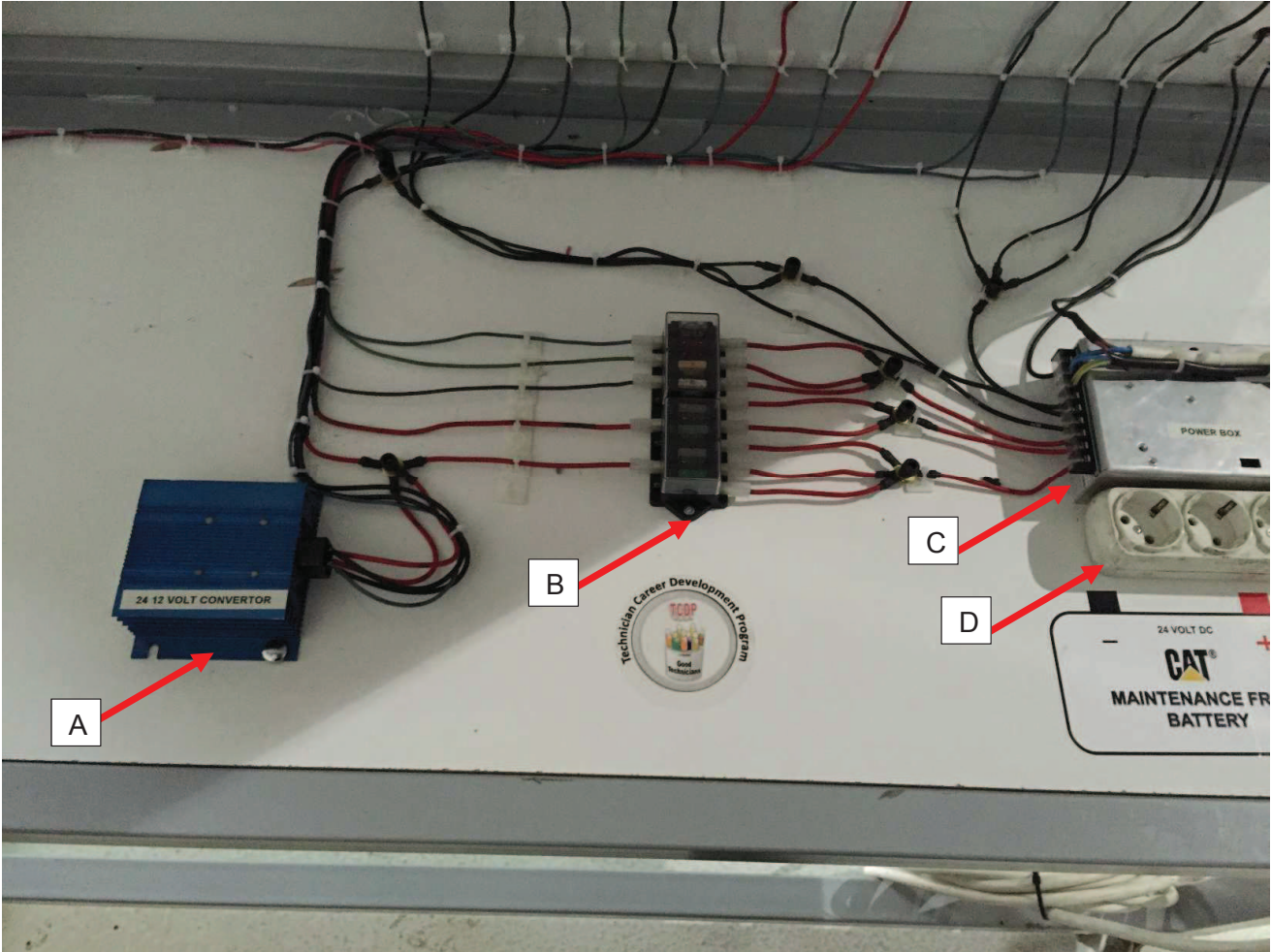
- A) Ac Control Panel
- B) Faster Speed Blower Relay
- C) Blower Motor Resistor
- D) Emergency Stop
- E) Blower And Blower Guide
- F) Thermostat Switch
- G) Evaporator
- H) Accumulator

- I) High Pressure Switch
- O) Compressor Clutch Relay
- P) Compressor
- R) In line Dryer
- S) Low Pressure Switch
- T) Condenser
- U) Supply And Ground Wires
- V) Water Valve

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The next step is to put electrical power supply devices to rear side of board.



- A) 24 Volt Dc/ 12volt Dc Convertor: The Water valve is supplied by 12 volt
- B) Fuse Box
- C) 220 Volt AC to 24 Volt DC Convertor
- D) 220 Volt AC Three Pin Plug: The pin plug is necessary for running vacuum pump.

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4.0 Benefits

- ❖ Air Conditioning Training
- ❖ Demonstrate parts of AC sytems.
- ❖ How to work circulation of refrigerant.
- ❖ How to work compressor clutch.
- ❖ How to work speed of blower fan.
- ❖ How to work water valf.
- ❖ Ability to measure voltage of blower motor resistor,water valf,relays,and control panel of AC systems.
- ❖ Ability to make some OJT trainings.



A) Demonstrate how to vacuum operation



B) Demonstrate how to fill refrigerant



C) Demonstrate how to measure AC PANEL



D) Demonstrate how to work compressor clutch

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5.0 Resources Required

Item		Cost (USD)
Material		300
Cat Parts		3000
Labor	Design	100
	Assembly	1000
Total		4400 \$

6.0 Supporting Attachments / References

The detail and drawings of AC training board is attachment
The electrical schematic of AC board is attachment

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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